

Elias Mawa

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Summary

Full-Stack Engineer specializing in distributed, event-driven SaaS platforms. Proven track record designing queue-based workflows, high-throughput APIs, and data-intensive services processing 100,000+ daily state transactions with strong consistency guarantees. Expert in Go, Python, and TypeScript, with a focus on domain-driven design, infrastructure automation, and production reliability at scale.

Professional Experience

Paireeo Inc.

Edmonton, AB | 2025 to Present

Lead Software Developer

- Architected a multi-tenant SaaS inventory platform in Go/TypeScript, implementing event-driven workflows to process 6,000+ requests per hour with strong consistency guarantees and <0.05% error rate across multi-step operations.
- Architected automated cloud infrastructure in DigitalOcean and CI/CD pipelines using GitHub Actions, containerized services with Docker to enable zero downtime deployments, and streamline release workflows.
- Managed end-to-end client support, prioritizing user-driven enhancements that impact the user experience and core platform functionality.
- Translated stakeholder requirements into domain-driven architecture, defining service boundaries and transactional guarantees for production.

First Canadian Financial Group

Sherwood Park, AB | 2023 to 2025

Full-Stack Software Developer

- Modernized legacy Perl modules by migrating logic to TypeScript and implementing RabbitMQ-backed async queues, decoupling long-running tasks and reducing peak-hour system load.
- Directed legacy modernization by reverse-engineering undocumented systems and migrating SOAP endpoints to RESTful APIs with strict validation, producing refactoring roadmaps that enabled secure partner integrations.
- Built automated Python reporting tools that replaced manual data aggregation, streamlining audit preparation and compliance tracking.
- Maintained mission-critical finance and insurance systems, ensuring strict compliance with provincial, federal, and industry regulations.

Chesney Management Group LTD.

Calgary, AB | 2021 to 2023

Full-Stack Software Developer

- Developed an internal ERP system to centralize inventory tracking, e-commerce operations, and business reporting.
- Replaced legacy PHP applications with a TypeScript, Next.js, and PostgreSQL stack, modernizing core web services and improving maintainability.
- Provisioned AWS and DigitalOcean infrastructure, containerizing services with Docker and building CI/CD pipelines to automate deployments.
- Built Python, Bash, and VBA automation scripts for purchase orders, payroll, and accounts payable, saving 80+ hours per week in manual processing and accelerating financial workflows.

Technical Skills

Languages: JavaScript/TypeScript, Bash, Python, Go, Dart, Java, PHP, Perl

Frontend: TypeScript (NEXT.js, Remix), Tailwind CSS, Flutter, Server-Side Rendering (SSR), State Management

Backend: Go (gin, echo, encore), TypeScript (Express, Koa, NestJS), PHP (Laravel), Python (Django, Flask)

Databases: PostgreSQL, MySQL, DB2 for I, MongoDB, Redis, Neo4j, Schema Design

Cloud & DevOps: Docker, Amazon Web Services (AWS), DigitalOcean, Google Cloud Platform (GCP), OpenTelemetry (OTEL), CI/CD

Architecture: Distributed Systems, Event-Driven Architecture, SaaS

AI/ML: AMD ROCm, Retrieval-Augmented Generation (RAG), Computer Vision

Projects

Grove | [GitHub](#)

2025

- Engineered an MCP-compatible agentic memory service to maintain context continuity across multi-tool AI sessions.
- Implemented a RAG pipeline using Qdrant for low-latency semantic retrieval and persistent cross-agent state management.
- Stack — Go, Qdrant

Fitr | [GitHub](#)

2024 — Present

Founding Developer

- Co-developed an AI-powered wardrobe management platform, architecting the core backend and API layer in Go.
- Built computer vision pipelines for clothing segmentation and pgvector-based semantic matching for personalized outfit recommendations.
- Optimized local LLM inference and embedding generation using AMD ROCm for hardware-accelerated model testing and development.
- Built with — Go, Flutter/Dart, PostgreSQL, pgvector, Redis

Burrow | [GitHub](#)

2024

- Developed a self-hosted reverse proxy alternative to Ngrok for secure local service exposure via HTTPS subdomains and TCP tunnels.
- Implemented dynamic routing and connection multiplexing in Go to support webhook testing and remote development workflows.
- Designed Redis-backed session caching and tunnel tracking to enable fast reconnections and low-latency state resolution.
- Built with — Go, Ngix, Redis

Projects Cont.

InkByte | [GitHub](#)

2021

- Built an end-to-end link shortener using Next.js, TypeScript, and Koa.js.
- Implemented server-side rendering, REST APIs, and database-driven routing.
- Built with — TypeScript, NEXT.JS, Postgres

GoBox | [GitHub](#)

2017

- Developed a file upload and sharing service using Go and a JavaScript frontend.
- Implemented HTTP servers, multipart uploads, and filesystem-based storage.
- Explored foundational backend concepts, including request handling and file I/O.
- Built with — Go, JavaScript, jQuery

Leadership & Community

Calgary Hornets Rugby Club

Calgary, AB | 2022 to 2023

VP of Marketing

- Managed online, social media, and print media channels for my local rugby club. Leveraged graphic design, and video editing skills to create visually stunning illustrations and content.
- Collaborated with executive members to align marketing efforts with club objectives. Developing marketing strategies to improve outreach and engagement of communication.
- Assisted the development of a revitalized website using PHP, and WordPress.
- Demonstrated leadership, communication, and cross-functional collaboration skills applied in professional engineering roles.

South Sudanese Youth of Canada

Calgary, AB | 2021 to 2023

Marketing Assistant / Event Organizer

- Developed marketing materials and campaigns for community events across digital and print channels.
- Assisted in organizing and coordinating events across Canada, supporting community engagement initiatives.
- Designed and delivered youth-focused workshops on computer science and technology topics.

The King's University Computer Science Club

Edmonton, AB | 2019 to 2021

President

- Led and organized technical workshops, presentations, and mentorship sessions for students.
- Invited and coordinated industry speakers to provide career insights and networking opportunities.
- Fostered a collaborative learning environment to improve engagement and participation in computer science.

The King's University Games Club

Edmonton, AB | 2018 to 2021

Event Organizer

- Organized and facilitated weekly events and gaming tournaments, fostering community participation.
- Coordinated logistics and programming for tabletop and video game events.

Education

The King's University, Edmonton, AB

B.Sc. Computer Science

Minor Communication Arts